

Transformer Architecture for 80M Parameter Chatbot

Sahil Ghelani

February 24, 2026

1 Overview

The Transformer is a neural network architecture based entirely on attention mechanisms. It consists of stacked encoder or decoder blocks.

Each block contains:

- Multi-Head Self Attention
 - Feed Forward Neural Network
 - Layer Normalization
 - Residual Connections
-

2 Attention Mechanism

Attention is computed as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

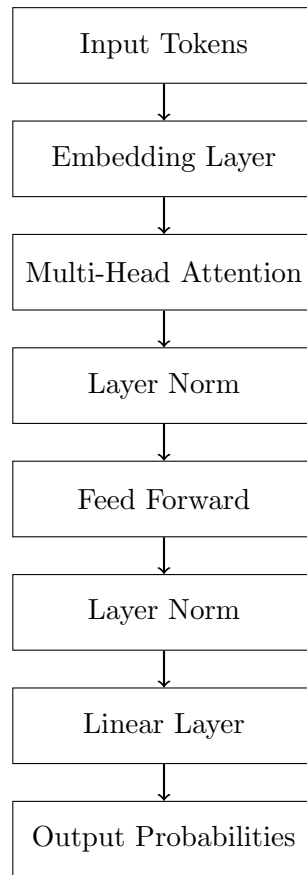
Where:

- Q = Query
 - K = Key
 - V = Value
 - d_k = dimension of key
-

3 Feed Forward Network

$$FFN(x) = max(0, xW_1 + b_1)W_2 + b_2$$

4 Transformer Block Diagram



5 Example Configuration (80M Parameters)

- Layers: 8
- Hidden Size: 512
- Attention Heads: 8
- Vocabulary Size: 30000
- Context Length: 256

6 Parameter Count Approximation

Embedding:

$$30000 \times 512 = 15.3M$$

Transformer Layers:

$$8 \times 7M = 56M$$

Output Layer:

$$512 \times 30000 = 15M$$

Total:

$$\approx 80M$$

—

7 Conclusion

This architecture is suitable for GPU-limited environments and chatbot applications.